IT Workshop II (Cloud Computing): Cloud Concepts and Technologies, Python for Cloud

August 6, 2026

Deployment

- **Deployment Design:** Application deployment is created with various tiers as specified in the deployment configuration. The variables in this step include the number of servers in each tier, computing, memory and storage capacities of server etc.
- **Performance Evaluation:** Once the application is deployed in the cloud, the next step in the deployment lifecycle is to verify whether the application meets the performance requirements with the deployment.
  - Monitoring the workload on the application.
  - Measuring various workload parameters such as response time and throughput.

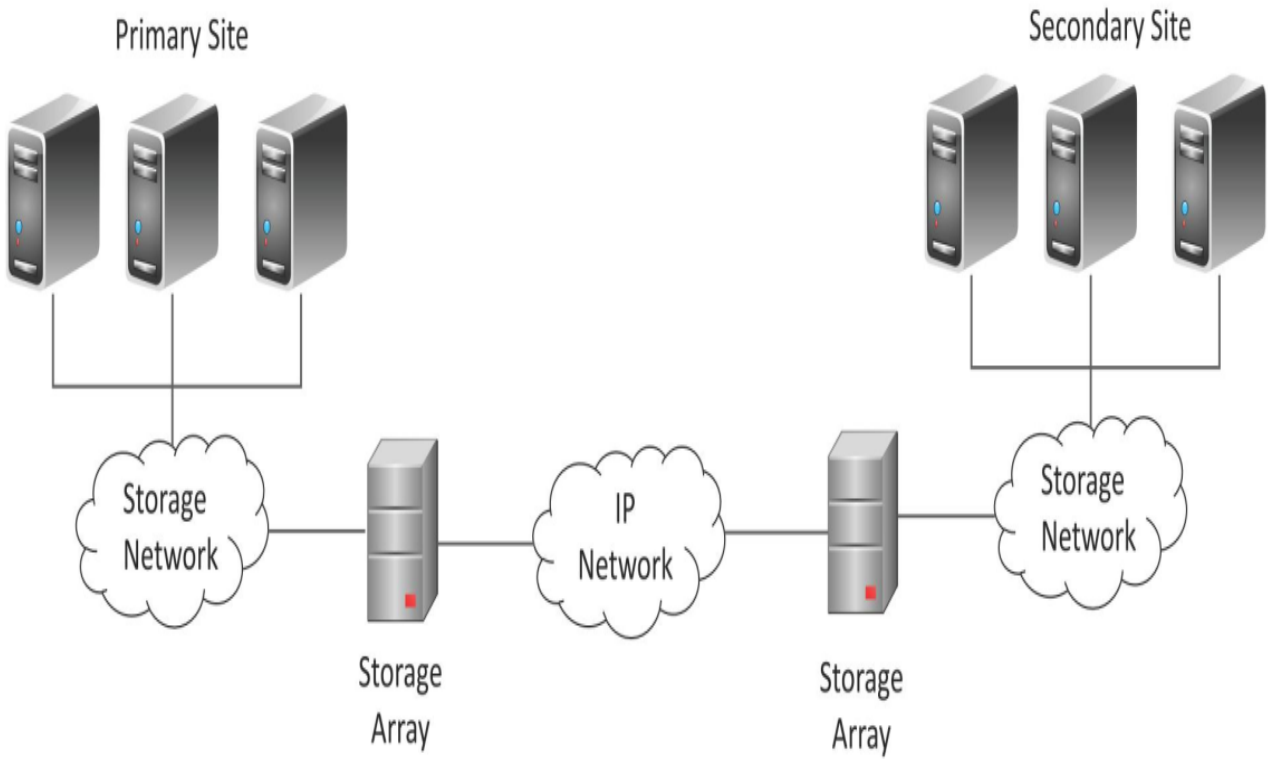
Deployment

Deployment Refinement: After evaluating the performance of the application, deployments are refined so that the application can meet the performance requirements. Various alternatives can exist in this step such as vertical scaling (or scaling up), horizontal scaling (or scaling out), alternative server interconnections, alternative load balancing etc.

Replication

- It is used to create and maintain multiple copies of the data in the cloud.
- Replication of data is important for practical reasons such as business continuity and disaster recovery.
- In the event of data loss at the primary location, organizations can continue to operate their applications from secondary data sources.
- With real-time replication of data, organizations can achieve faster recovery from failures.

Replication



Array-based Replication

Monitoring

- Cloud resources can be monitored by monitoring services provided by the cloud service providers.
- **Monitoring Services:**
  - Collects data on various systems and application metrics from the cloud computing instances.
  - Users can define various actions based on the monitoring data, for example, auto-scaling a cloud deployment when the CPU usage of monitored resources becomes high.
  - Monitoring services also provide various statistics based on the monitoring data collected.

Monitoring

Examples of Monitoring Metrics

Type	Metrics
CPU	CPU-Usage, CPU-Idle
Disk	Disk-Usage, Bytes/sec (read/write), Operations/sec
Memory	Memory-Used, Memory-Free, Page-Cache
Interface	Packets/sec (incoming/outgoing), Octets/sec(incoming/outgoing)

Amazon Elastic Compute Cloud

- Amazon Elastic Compute Cloud (EC2) is a compute service provided by Amazon.
- Keypairs are used to securely connect to an instance after it launches.
- Security groups are used to open or block a specific network port for the launched instances.
- The public DNS can be used to securely connect to the instance using SSH.

Amazon Simple Storage Services

- Amazon Simple Storage Service(S3) is an online cloud-based data storage infrastructure for storing and retrieving any amount of data.
- Data stored on S3 is organized in the form of buckets. You must create a bucket before you can store data on S3.
- We can upload files to S3.
- While uploading a file, we can specify the redundancy and encryption options and access permissions.

Identity and Access Management

- Identity and Access Management (IDAM) for cloud describes the authentication and authorization of users to provide secure access to cloud resources.
- IDAM services allow organizations to manage users, access permissions, security credentials and access keys.
- Organizations can enable role-based access control to cloud resources and applications.
- IDAM services allow creation of user groups where all the users in a group have the same access permissions.

The End